

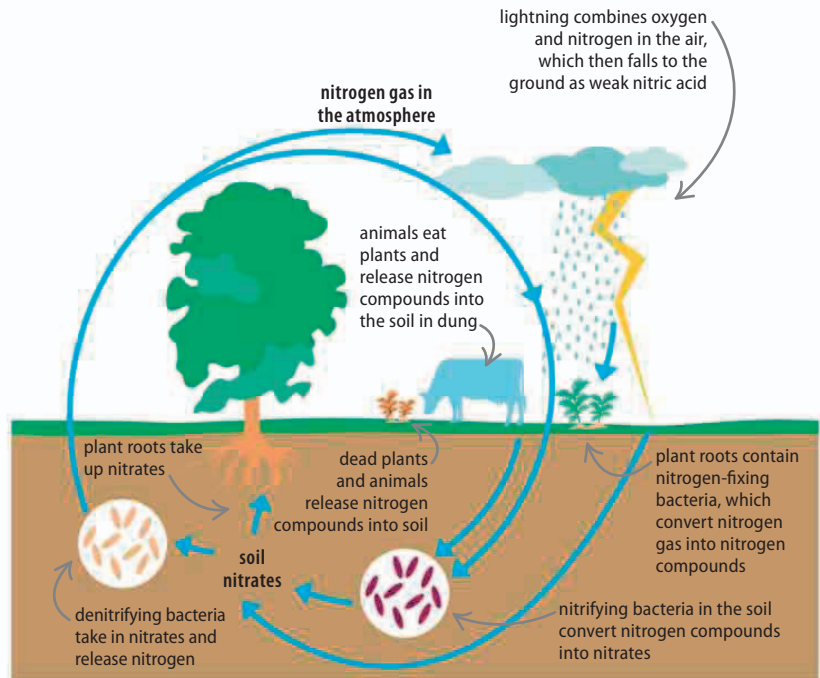
The nitrogen cycle

Nitrogen is an essential component in amino acids, the basic units of protein, which all living creatures need. Therefore, all life needs a supply of nitrogen compounds. Animals cannot manufacture most amino acids themselves, so they obtain them from plant foods. Plants make amino acids from nitrates (a combination of nitrogen and oxygen) absorbed from the soil. The nitrates are added to the soil by bacteria that fix nitrogen from the air.

REAL WORLD

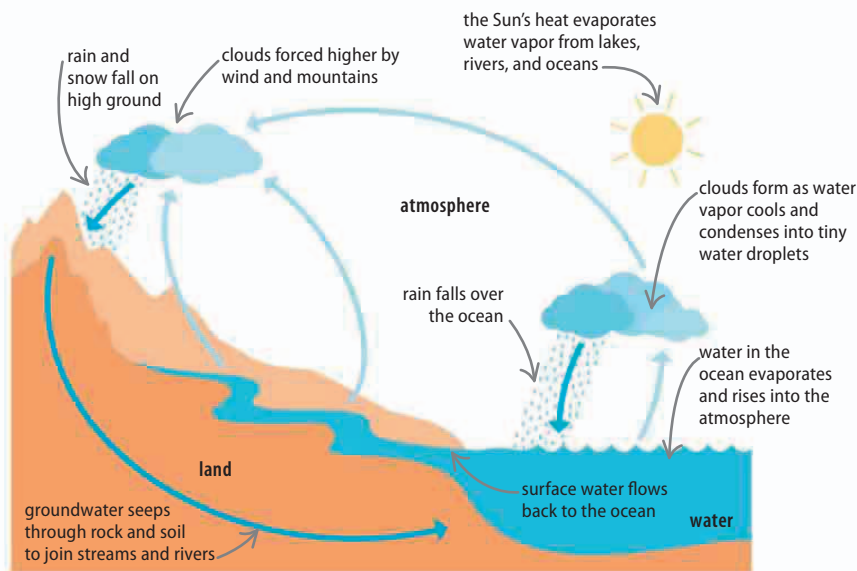
Carnivorous plant

The Venus flytrap grows in soils that lack nitrates, so the plant collects it from prey instead. It traps insects in its pressure-sensitive, pincer-shaped leaves. The leaves shut to form a stomachlike space, where enzymes digest the insect to release its nutrients.



△ Nitrates

Nitrogen is not a very reactive element, mostly staying unchanged in the atmosphere. However, the enzymes in certain bacteria and the high energy of lightning can convert nitrogen into nitrates, a form that can be used by all life.



The water cycle

Earth's water is always on the move, collecting in vast quantities in the ocean, but rarely finding its way to deserts. Life cannot exist without water. It is one of the ingredients in the production of glucose in photosynthesis, and water is also the medium in which the metabolic process takes place inside cells. Most living bodies are mainly water—about 60 percent in the case of humans—and, where water is rare, so is life.

◁ Movement of water

Most of Earth's water is in the oceans, but it also moves constantly into the atmosphere, falling as rain to form freshwater running over and into the ground or freezing as ice on high mountains and in the polar regions.